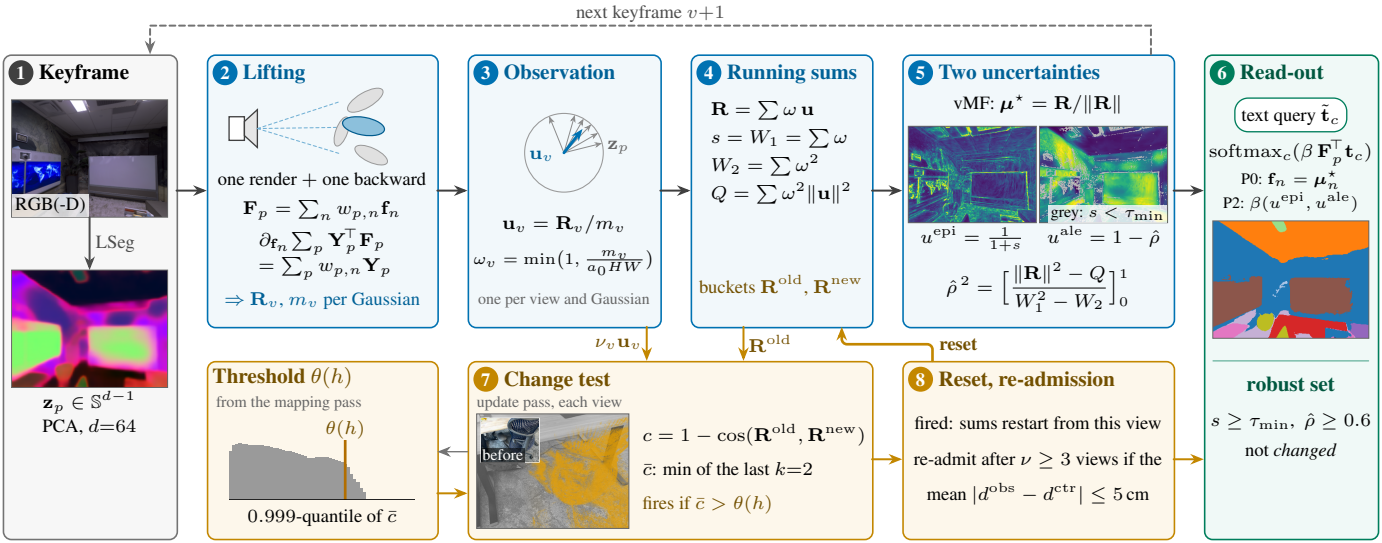

next keyframe $v+1$
1 Keyframe
RGB(-D)
LSeg
$\mathbf{z}_p \in \mathbb{S}^{d-1}$
PCA, d=64
2 Lifting
one render + one backward
$\mathbf{F}_p = \sum_n w_{p,n} \mathbf{f}_n$
$\partial_{\mathbf{f}_n} \sum_p \mathbf{Y}_p^\top \mathbf{F}_p$
$= \sum_p w_{p,n} \mathbf{Y}_p$
$\Rightarrow \mathbf{R}_v, m_v$ per Gaussian
3 Observation
$\mathbf{u}_v$
$\mathbf{z}_p$
$\mathbf{u}_v = \mathbf{R}_v / m_v$
$\omega_v = \min(1, \frac{m_v}{a_0 HW})$
one per view and Gaussian
4 Running sums
$\mathbf{R} = \sum \omega \mathbf{u}$
$s = W_1 = \sum \omega$
$W_2 = \sum \omega^2$
$Q = \sum \omega^2 \|\mathbf{u}\|^2$
buckets $\mathbf{R}^{\text{old}}$, $\mathbf{R}^{\text{new}}$
5 Two uncertainties
vMF: $\boldsymbol{\mu}^\star = \mathbf{R}/\|\mathbf{R}\|$
grey: $s < \tau_{\min}$
$u^{\text{epi}} = \frac{1}{1+s}$
$u^{\text{ale}} = 1 - \hat{\rho}$
$\hat{\rho}^2 = \left[\frac{\|\mathbf{R}\|^2 - Q}{W_1^2 - W_2}\right]_0^1$
6 Read-out
text query $\tilde{\mathbf{t}}_c$
$\text{softmax}_c(\beta\, \mathbf{F}_p^\top \mathbf{t}_c)$
P0: $\mathbf{f}_n = \boldsymbol{\mu}_n^\star$
P2: $\beta(u^{\text{epi}}, u^{\text{ale}})$
robust set
$s \geq \tau_{\min}$, $\hat{\rho} \geq 0.6$
not *changed*
$\nu_v \mathbf{u}_v$
$\mathbf{R}^{\text{old}}$
reset
Threshold $\theta(h)$
from the mapping pass
$\theta(h)$
0.999-quantile of $\bar{c}$
7 Change test
update pass, each view
before
$c = 1 - \cos(\mathbf{R}^{\text{old}}, \mathbf{R}^{\text{new}})$
$\bar{c}$: min of the last k=2
fires if $\bar{c} > \theta(h)$
8 Reset, re-admission
fired: sums restart from this view
re-admit after $\nu \geq 3$ views if the
mean $|d^{\text{obs}} - d^{\text{ctr}}| \leq 5$ cm